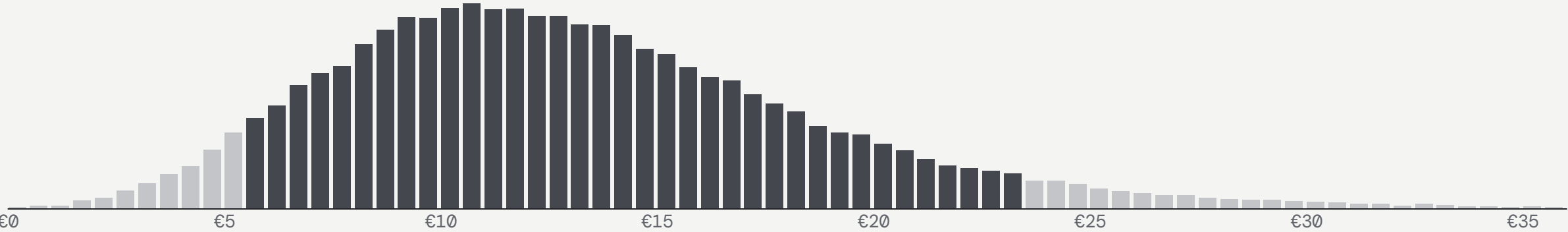


What it is worth, and how likely each value is.

Lumina Renewables is a fictional Italian solar and wind operator. Instead of one share price, six assumptions become probability distributions and the DCF runs 50,000 times.

VALUE PER SHARE, 50,000 RUNS



BASE REVENUE	EBITDA MARGIN	NET DEBT	SHARES	MARKET PRICE	BASE CASE
€145m	42%	€185m	25m	€11.50	€14.92

SIMULATED MEDIAN

€12.49

the typical outcome, per share

P5 TO P95

€5.35 to €23.61

nine runs in ten fall inside

VALUE ABOVE MARKET

57.5%

share of runs above €11.50

WACC SHARE OF VARIANCE

38%

the largest single driver

Six guesses, each drawn as a distribution.

DRIVER	SHAPE	MIN • MODE • MAX	RANK CORR. WITH VALUE	SHARE OF VARIANCE
Revenue growth		4% • 9% • 14%	+0.42	17%
EBITDA margin		36% • 42% • 46%	+0.44	18%
WACC		8.5% ± 1.1%	-0.63	38%
Tax rate		24% • 26% • 35%	-0.15	2%
Terminal growth		1% • 2% • 2.75%	+0.16	2%
CapEx (% of revenue)		12% • 16% • 22%	-0.47	22%

ITERATIONS

50,000

GROWTH-MARGIN CORRELATION

+0.35

WACC MINUS G FLOOR

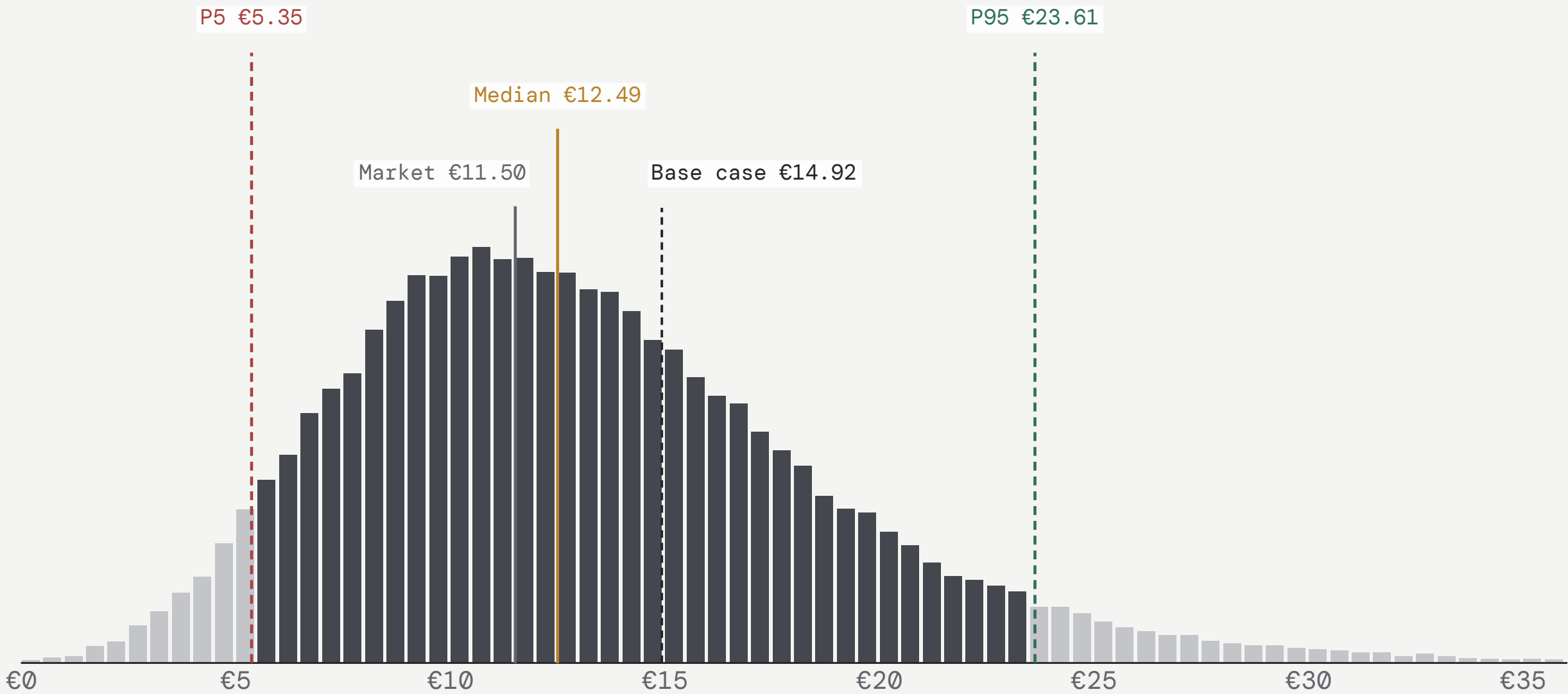
1.50 pts

HORIZON

5 yrs + TV

A distribution, not a number.

EQUITY VALUE PER SHARE, 50,000 RUNS • DARKER BARS: THE MIDDLE 90%



P 5 • DOWNSIDE

€5.35

one run in twenty is worse

MEDIAN

€12.49

the mean is €13.23

P 95 • UPSIDE

€23.61

one run in twenty is better

VALUE ABOVE €11.50

57.5%

the odds of being cheap today

The base case sits 19.5% above the typical outcome.

A most-likely input does not give the most-likely output when the distributions are lopsided. Switching on one uncertainty at a time, on the same random numbers, shows where the €2.43 goes.

TAX, CAPEX AND MARGIN TAILS

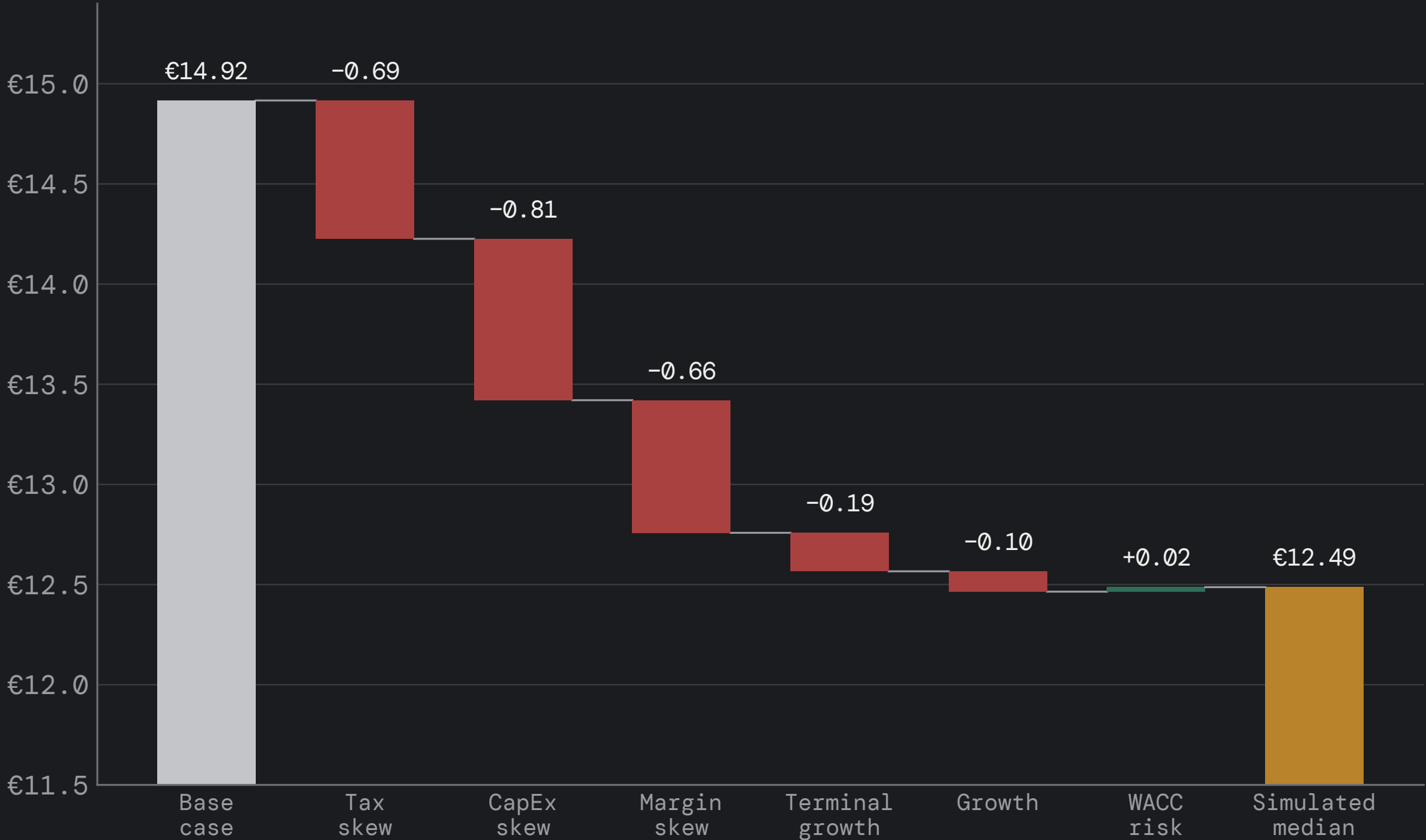
-€2.16

three skews against value

WACC RISK

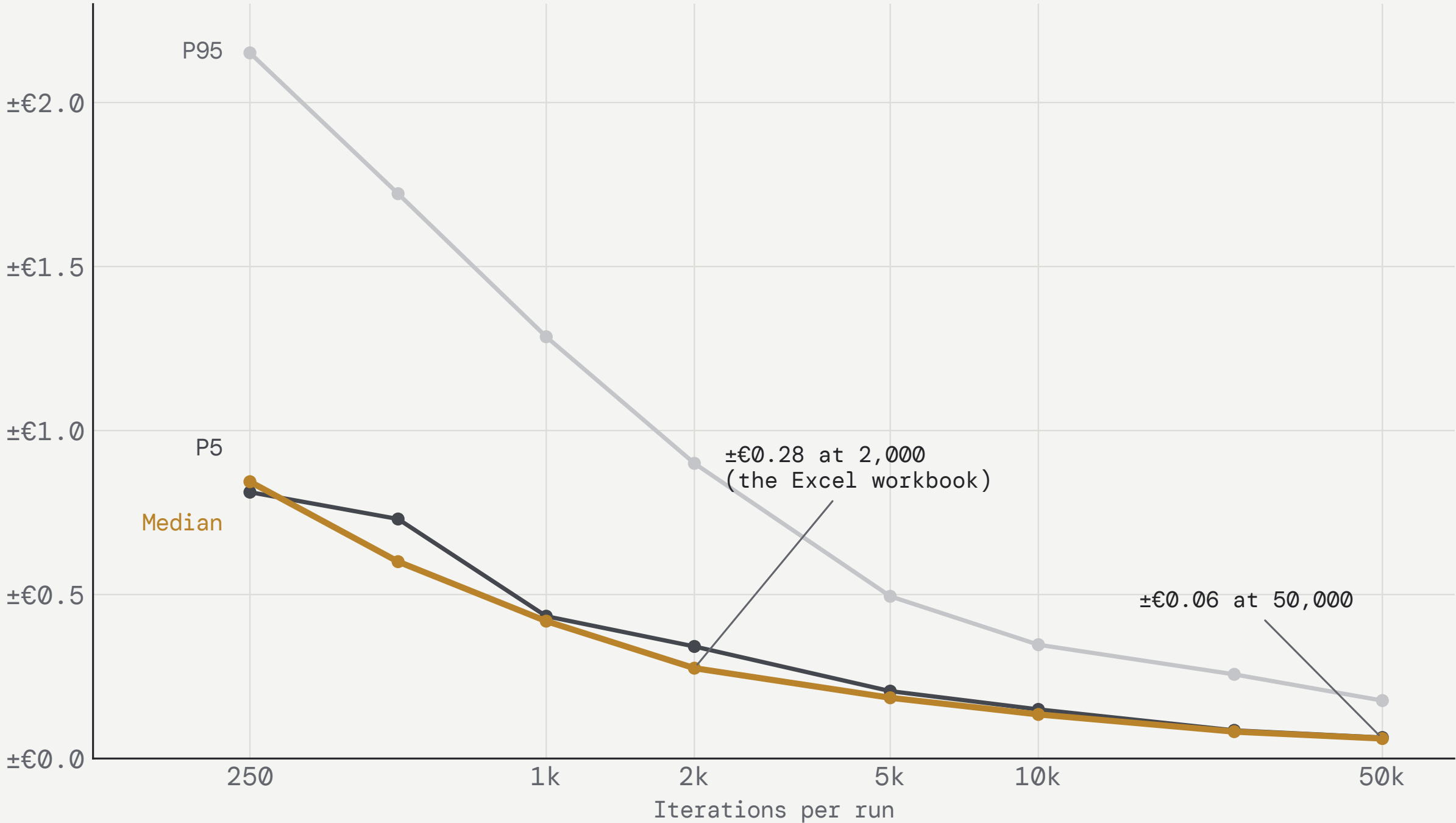
+0.02

symmetric: it widens the range, not the centre



2,000 runs pin the median; the tails need 25,000.

95% RANGE OF EACH STATISTIC ACROSS 200 FULL RE-RUNS



MEDIAN AT 2,000 RUNS

±€0.28
the size of the Excel workbook

MEDIAN AT 50,000

±€0.06
the headline run

P95 AT 2,000 RUNS

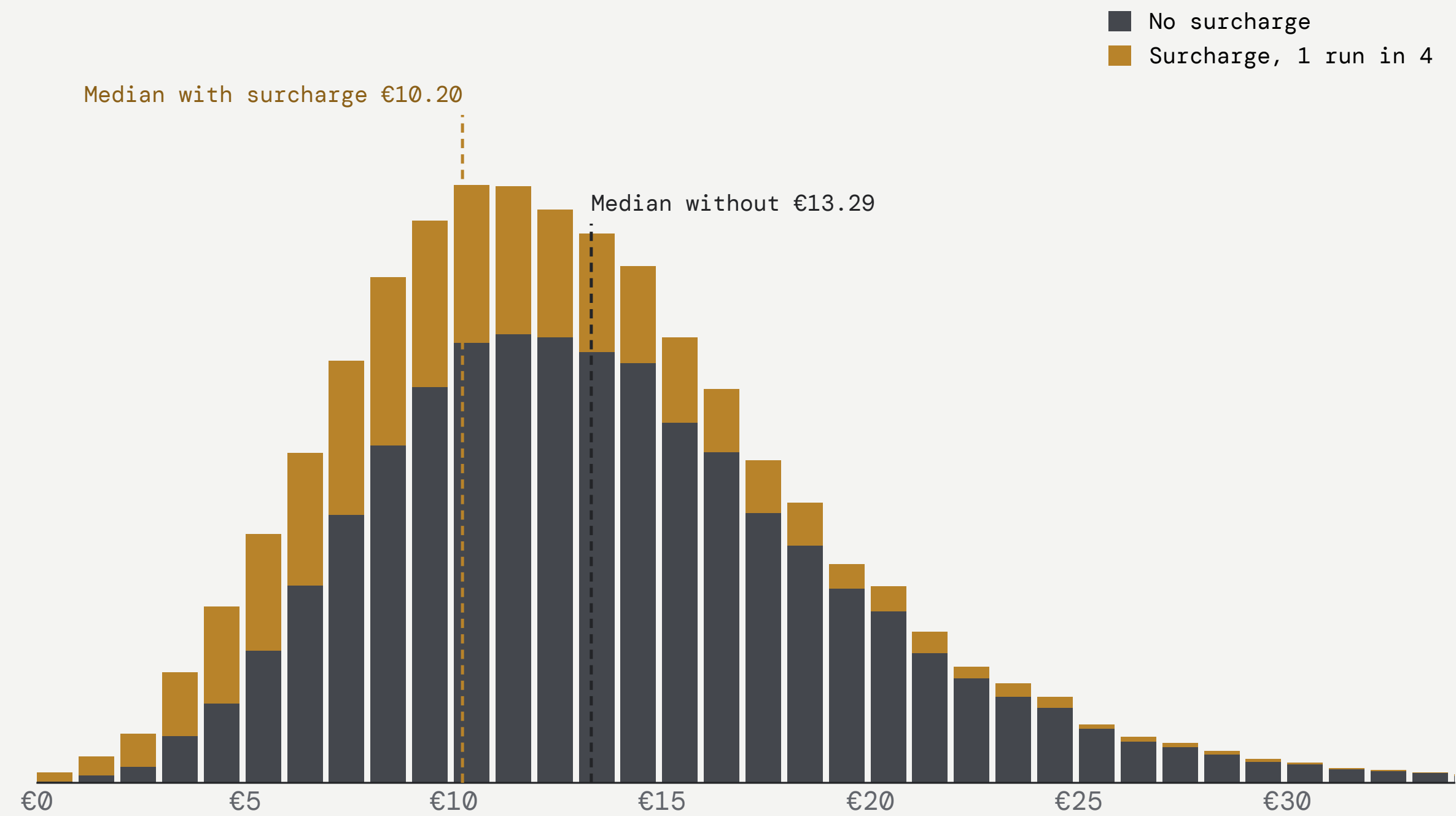
±€0.90
tails converge more slowly

PYTHON VS EXCEL

€0.00
all 2,000 values reproduced

One political decision is worth €3.09 a share.

VALUE PER SHARE: 9-POINT WINDFALL SURCHARGE WITH A 25% PROBABILITY, SAME RANDOM NUMBERS



MEDIAN, NO SURCHARGE

€13.29

value above price in 64% of runs

MEDIAN, SURCHARGE

€10.20

value above price in 40% of runs

HEADLINE MEDIAN

€12.53

vs €12.49 with a smooth tax tail

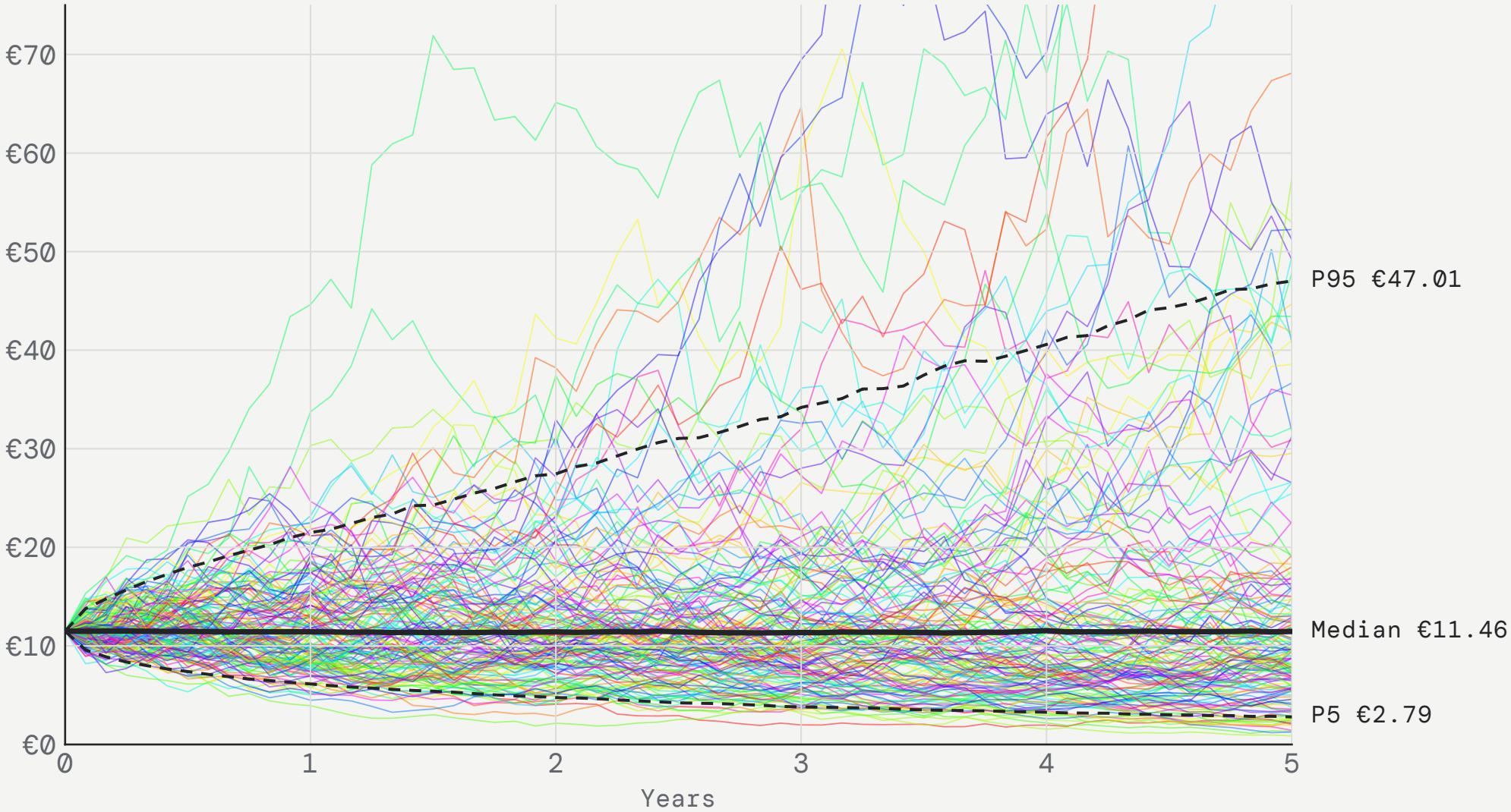
AVERAGE TAX

28.3%

same in both models

Cheap on the median too, but less than the base case says.

SHARE PRICE, 5 YEARS OF GBM PATHS • DRIFT 7%, VOLATILITY 38%



UPSIDE AT €11.50

8.6%

on the median; 29.7% on the base case

WHAT TO PRICE

WACC • tax

38% of variance • €3.09 a share

DELIVERABLES

Notebook • Excel

memo • deck • live simulator

LIMITS

Judgement

Lumina is fictional; the ranges are illustrative, not market data